

Discrete Math W2

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Hello

I am Diego

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DM HS2026 - Group A

WhatsApp group



Plan

1. Feedback
2. Theory Recap
3. Exercises
4. Questions?

Feedback

Theory Recap

Logical Operators / Formulas

A	B	$\neg A$	$A \wedge B$	$A \vee B$	$A \rightarrow B$	$A \leftrightarrow B$
0	0	1	0	0	1	1
0	1	1	0	1	1	0
1	0	0	0	1	0	0
1	1	0	1	1	1	1

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Definition 2.5

A correctly formed expression involving the propositional symbols $A, B, C, \dots$ and logical operators is called a **formula**.

Definition 2.6

Two formulas F and G (in propositional logic) are called **equivalent**, denoted as $F \equiv G$, if they correspond to the same function, i.e., if the truth values are equal for all truth assignments to the propositional symbols appearing in F and G .

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Definition 2.7

A formula G is a **logical consequence** of a formula F , denoted

$$F \models G$$

if for all truth assignments to the propositional symbols appearing in F or G , the truth value of G is 1 if the truth value of F is 1.

Definition 2.8

A formula F (in propositional logic) is called a **tautology** or **valid** if it is true for all truth assignments of the involved propositional symbols. One often writes $\models F$ to say that F is a tautology.

Definition 2.9

A formula F (in propositional logic) is called **satisfiable** if it is true for at least one truth assignment of the involved propositional symbols, and it is called **unsatisfiable** otherwise.

Some useful lemmas

Lemma 2.2

A formula F is a tautology if and only if $\neg F$ is unsatisfiable.

Lemma 2.3

For any formulas F and G , $F \rightarrow G$ is a tautology if and only if $F \models G$

Why do we need predicate logic?

Propositional logic is not expressive enough.

We want to be able to express more statements in a formal way.

For this we need an extension to propositional logic.

Universe

A universe is a non-empty set U .

Universe / Predicate

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Definition 2.10

A k -ary **predicate** P on U is a function $U^k \rightarrow \{0, 1\}$

Let $U = \mathbb{Q}$

$$\text{prime}(x) = \begin{cases} 1 & \text{if } x \text{ is prime} \\ 0 & \text{else} \end{cases} \quad D(x, y) = \begin{cases} 1 & y \neq x \\ 0 & \text{else} \end{cases}$$

$$Q(x, y) = 1 \iff x \text{ and } y \text{ have the same denominator}$$

You can use infix notation

$$x < y : \text{IsLessThan}(x, y)$$

$$\leq \geq \cdot =$$

Definition 2.11

For a universe U and a predicate $P(x)$, we define the following logical statements

$\forall x P(x)$ stands for: $P(x)$ is true for **all** $x \in U$

$\exists x P(x)$ stands for: $P(x)$ is true for **some** $x \in U$

Quantifiers

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- $U = \mathbb{N}$: $\forall x (x \geq 0)$
- $U = \mathbb{N}$: $\forall x \exists y (x + y = 0)$
- $U = \mathbb{Q}$: $\forall x \exists y (x + y = 0)$
- $U = \mathbb{R}$: $\forall x (\exists y (y^2 = x) \rightarrow (x \geq 0))$
- $U = \mathbb{N} \setminus \{0\}$: $\exists x \exists y \exists z (x^2 + y^2 = z^2)$
- $U = \mathbb{N}$: $\exists y \forall x (x \leq y)$

Interpretation of Formulas

Each formula has parts that are left open for interpretation

Universe, Predicates

Depending on those, the formula can either be true or false

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Without fixing an interpretation, a formula is not a mathematical statement!

More in Chapter 6.

Some short exercises

1) Let $U = \mathbb{R}$. What is the meaning of the following formula and is it true or false?

$$\forall x (x = 0 \vee \exists y (xy = 1))$$

2) What is the meaning of the following formula, and for which universes is it true (or false)?

$$\forall x \forall y ((x < y) \rightarrow \exists z ((x < z) \wedge (z < y)))$$

3) What is the meaning of the following formula, and for which universes is it true (or false)?

$$\exists x \forall y (x \leq y)$$

4) Let $U = \mathbb{R}$. What is the meaning of the following formula and is it true or false?

$$\forall x ((x \geq 0) \rightarrow \exists y (y^2 = x))$$

Informal Definition

A formula in predicate logic is **satisfiable** if there is an interpretation of the involved symbols that makes the formula true.

Satisfiability / Tautologies (Valid)

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A formula in predicate logic is **satisfiable** if there is an interpretation of the involved symbols that makes the formula true.

Informal Definition

A formula in predicate logic is a **tautology** if it is true for all choices of the universe and for all interpretations of the predicates.

More details in Chapter 6

See exercise 4 on my exercise sheet for a fun exercise.

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Same definitions and rules as for propositional logic.

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- $\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$
→ *Only holds for identical quantifiers*
- $\forall x (x = 0 \vee \exists y (xy = 1)) \equiv \forall x (\neg(x = 0) \rightarrow \exists y (xy = 1))$

Useful Rules for Predicate Logic

- $\forall x P(x) \wedge \forall x Q(x) \equiv \forall x (P(x) \wedge Q(x))$

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 $\exists x P(x) \wedge \exists x Q(x) \not\models \exists x (P(x) \wedge Q(x))$

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→ *Note: The converse does not hold:*
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- $\neg \forall x P(x) \equiv \exists x \neg P(x)$
- $\neg \exists x P(x) \equiv \forall x \neg P(x)$

Useful Rules for Predicate Logic

- $\forall x P(x) \wedge \forall x Q(x) \equiv \forall x (P(x) \wedge Q(x))$
- $\exists x (P(x) \wedge Q(x)) \models \exists x P(x) \wedge \exists x Q(x)$
→ *Note: The converse does not hold:*
 $\exists x P(x) \wedge \exists x Q(x) \not\models \exists x (P(x) \wedge Q(x))$
- $\neg \forall x P(x) \equiv \exists x \neg P(x)$
- $\neg \exists x P(x) \equiv \forall x \neg P(x)$
- $\exists y \forall x P(x, y) \models \forall x \exists y P(x, y)$

Useful Rules for Predicate Logic

- $\forall x P(x) \wedge \forall x Q(x) \equiv \forall x (P(x) \wedge Q(x))$
- $\exists x (P(x) \wedge Q(x)) \models \exists x P(x) \wedge \exists x Q(x)$
→ *Note: The converse does not hold:*
 $\exists x P(x) \wedge \exists x Q(x) \not\models \exists x (P(x) \wedge Q(x))$
- $\neg \forall x P(x) \equiv \exists x \neg P(x)$
- $\neg \exists x P(x) \equiv \forall x \neg P(x)$
- $\exists y \forall x P(x, y) \models \forall x \exists y P(x, y)$
- **BUT:** $\forall x \exists y P(x, y) \not\models \exists y \forall x P(x, y)$

Logical Formulas vs Mathematical Statements

In general, a logical formula is **NOT** a mathematical statement.
Different interpretations lead to different mathematical statements.
Without fixing an interpretation, a formula is not a mathematical statement!

Mathematical statements about formulas

- F is valid
- F is unsatisfiable
- $F \models G$

Examples

Mathematical statements about formulas

- F is valid
- F is unsatisfiable
- $F \models G$

$F \text{ is valid} \implies G \text{ is valid} \quad (*)$

$\neq$

$F \models G$

Examples

Mathematical statements about formulas

- F is valid
- F is unsatisfiable
- $F \models G$

$$\begin{array}{ccc} F \text{ is valid} & \implies & G \text{ is valid} & (*) \\ & \neq & \\ & & F \models G \end{array}$$

Lemma 2.4

For any two formulas F and G , if $F \models G$, then $(*)$ is true.

Exercises

Exercise 2.6 Short Questions (Old Exam Questions)

This exercise contains several short questions from exams from previous years.

1. Find two propositional formulas F and G , such that $F \rightarrow G$ is unsatisfiable.
2. The statement " $F \wedge G$ is satisfiable if and only if F is satisfiable and G is satisfiable" is true for any formulas F and G . True or False?
3. A formula F in propositional logic is unsatisfiable if and only if $\neg F$ is a tautology. True or False?
4. The following formula is a tautology: $A \rightarrow (\neg A \rightarrow B)$. True or False?

Logical Consequence

Prove or disprove the following statements about formulas.

1. $\neg B \wedge (A \rightarrow B) \models \neg A$
2. $A \vee B \models A \wedge B$
3. $\models (A \wedge (A \rightarrow B)) \rightarrow B$
4. $A \rightarrow B \models \neg B \rightarrow \neg A$

Satisfiability and Tautologies

For each of the formulas below, determine whether it is satisfiable or unsatisfiable and whether it is a tautology or not.

1. $(A \vee B) \rightarrow (A \wedge B)$
2. $(A \rightarrow B) \wedge (A \wedge \neg B)$
3. $((A \vee B) \wedge \neg A) \rightarrow B$
4. $A \leftrightarrow (B \wedge \neg A)$

Simplifying a Formula (Exam Task)

Simplify the following formula such that it contains at most one $\wedge$ and one $\vee$

$$F \equiv (A \wedge (B \rightarrow C)) \vee \neg((A \rightarrow \neg B) \rightarrow \neg C)$$

Lemma 6.1. *For any formulas F , G , and H we have*

- 1) $F \wedge F \equiv F$ and $F \vee F \equiv F$ (idempotence);
- 2) $F \wedge G \equiv G \wedge F$ and $F \vee G \equiv G \vee F$ (commutativity);
- 3) $(F \wedge G) \wedge H \equiv F \wedge (G \wedge H)$ and $(F \vee G) \vee H \equiv F \vee (G \vee H)$ (associativity);
- 4) $F \wedge (F \vee G) \equiv F$ and $F \vee (F \wedge G) \equiv F$ (absorption);
- 5) $F \wedge (G \vee H) \equiv (F \wedge G) \vee (F \wedge H)$ (distributive law);
- 6) $F \vee (G \wedge H) \equiv (F \vee G) \wedge (F \vee H)$ (distributive law);
- 7) $\neg\neg F \equiv F$ (double negation);
- 8) $\neg(F \wedge G) \equiv \neg F \vee \neg G$ and $\neg(F \vee G) \equiv \neg F \wedge \neg G$ (de Morgan's rules);
- 9) $F \vee \top \equiv \top$ and $F \wedge \top \equiv F$ (tautology rules);
- 10) $F \vee \perp \equiv F$ and $F \wedge \perp \equiv \perp$ (unsatisfiability rules).
- 11) $F \vee \neg F \equiv \top$ and $F \wedge \neg F \equiv \perp$.

Translate Statements to Logic

Let $U = \mathbb{Z}$

Translate the following statements into propositional logic. You may define your own sensible predicates.

1. Every integer has an inverse.
2. The square of every odd integer is odd.
3. For every integer $n > 1$, there exists at least one prime number p such that $n < p < 2n$.
4. There exists exactly one even prime number.
5. There are infinitely many primes p such that $p + 2$ is prime.
6. For every integer $k > 0$, there exist k consecutive composite integers.
7. There are infinitely many positive integers that cannot be written as the sum of three squares.

Describing Formulas

Let $U = \mathbb{Q}$

We define the following predicates

$$P(x) = \begin{cases} 1, & x \in \mathbb{Z} \\ 0, & \text{Else} \end{cases} \quad Q(x) = \begin{cases} 1, & x < 1 \\ 0, & \text{Else} \end{cases} \quad R(x, y, z) = \begin{cases} 1, & x + y = z \\ 0, & \text{Else} \end{cases}$$

Describe each of the formulas in words and decide if it is true or false.

1. $\exists x (P(x) \wedge Q(x))$
2. $\exists x (Q(x) \wedge \forall y R(x, y, y))$
3. $\forall x \forall y (R(x, x, y) \rightarrow P(y))$
4. $\forall x (Q(x) \rightarrow \exists y (Q(y) \wedge R(y, y, x)))$

Questions?